MINGYU KIM

CONTACT INFORMATION

Email: k0136000@mju.ac.kr
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RESEARCH INTEREST

My master's research focused on enhancing dynamic obstacle avoidance performance in mobile robot navigation by leveraging deep learning models to process semantic information and generate human-like trajectories based on imitation learning.

My primary research interests are focused on:

- Multimodal learning
- Learning-based path planning
- Vision-Language-Action model

EDUCATIONAL QUALIFICATION

Myongji University — Yongin, Korea
M.S. in Information and Communication Engineering — Mar. 2024 - Current
Total GPA of 4.5/4.5

Myongji University — Yongin, Korea
B.S. in Information and Communication Engineering — Mar. 2020 - Feb. 2024
Total GPA of 3.5/4.5, Major GPA of 3.77/4.5

PUBLICATIONS

International Journals

- **HADP: Hybrid A*-Diffusion Planner for Robust Navigation in Dynamic Obstacle Environments** paper
  M. Kim*, C. Heo*, J. Jung (*co-first authors)
  IEEE Access [Impact Factor: 3.6]
  overview
- **HiMSELF: A Hierarchical Misbehavior Classification with Sequence Embedding by Latent Features in Vehicular Ad-Hoc Networks** paper
  M. Kim, D. Yum, J. Jung
  IEEE Access [Impact Factor: 3.6]
  overview

Domestic Journals

- **Effective Embedding Techniques for Misbehavior Classification in Vehicular Ad-Hoc Networks** paper
  M. Kim, J. Jung
  Journal of Korean Institute of Information Scientists and Engineers, (KIISE 2024).

Domestic Conferences

- **Label Similarity Analysis Based on LSTM in Vehicular Ad-hoc Networks** paper
  M. Kim, J. Jung
  Korea Software Congress, (KSC 2023) – **Best Paper Award**.
- **Reinforcement Learning-Based Navigation System Without Dependence on SLAM** paper
  M. Kim, J. Jung
  Korea Institute of Next Generation Computing, (KINGPC 2024).

RESEARCH PROJECTS

Development of a Cybersecurity Platform for V2X-based Remote-Assisted Mobility
AutoCrypt — 2023 – Current

Semantic Autonomous Driving Systems Utilizing Diffusion Models for Enhanced Dynamic Obstacle Avoidance Performance
National Research Foundation of Korea (NRF) — 2024 – Current

PATENTS

Method for producing dynamic obstacle avoidance path using conditional diffusion model

Status:	Patent Application Filed
Application No.:	10-2025-0053197
Filing Date:	Apr 23, 2025
Applicants:	Jaehee Jung, **Mingyu Kim**, Chanyeong Heo

System for producing dynamic obstacle avoidance path using conditional diffusion model

Status:	Patent Application Filed
Application No.:	10-2025-0053198
Filing Date:	Apr 23, 2025
Applicants:	Jaehee Jung, **Mingyu Kim**, Chanyeong Heo

AWARDS

Encouragement Prize, Myongji University Software Competition — 2022

Grand Prize, Myongji University Software Competition — 2023

Best Paper Award, Korea Software Congress (KSC) — 2023

Gold Prize in the Contribution Award, Myongji University — 2024

Awarded the Research Scholarship for M.S. Studies by the National Research Foundation (NRF) — 2024

SKILLS

Programming Language

- Python (PyTroch, Tensorflow, FastAPI), C (MCU programming), JavaScript (NestJS)

Machine Learning

- Imitation learning, Anomaly Detection

Robot Development and Implementation

- Learning-Based Path Planning
- Development of a Swerve-Drive Robot
- Development of an Omni-Wheel-Based Robot

Development Tools

- ROS, Gazebo, Fusion 360

EXTRACURRICULAR ACTIVITIES:

Undergraduate Research Intern, DS LAB (Advisor: Jaehee Jung) 2022 – 2024

- Researched on improving misbehavior classification in V2X messages by analyzing relationships between misbehavior types.
- Researched on DRL-based navigation systems that operate without reliance on maps.

LikeLion, Myongji University 2023 – 2024

- Backend Operator & Instructor

AUTURBO 2024 – 2025

- Member, Outdoor Delivery Robot Team

PERSONAL INFORMATION:

Date of Birth	20-02-2001
Hobbies	3D Modeling Freelance work via Soomgo